

# A fluid-to-solid jamming transition underlies vertebrate body axis elongation

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**Just as in clay moulding or glass blowing, physically sculpting biological structures requires the constituent material to locally flow like a fluid while maintaining overall mechanical integrity like a solid. Disordered soft materials, such as foams, emulsions and colloidal suspensions, switch from fluid-like to solid-like behaviours at a jamming transition<sup>1–4</sup>. Similarly, cell collectives have been shown to display glassy dynamics in 2D and 3D<sup>5,6</sup> and jamming in cultured epithelial monolayers<sup>7,8</sup>, behaviours recently predicted theoretically<sup>9–11</sup> and proposed to influence asthma pathobiology<sup>8</sup> and tumour progression<sup>12</sup>. However, little is known about whether these seemingly universal behaviours occur in vivo<sup>13</sup> and, specifically, whether they play any functional part during embryonic morphogenesis. Here, by combining direct in vivo measurements of tissue mechanics with analysis of cellular dynamics, we show that during vertebrate body axis elongation, posterior tissues undergo a jamming transition from a fluid-like behaviour at the extending end, the mesodermal progenitor zone, to a solid-like behaviour in the presomitic mesoderm. We uncover an anteroposterior, N-cadherin-dependent gradient in yield stress that provides increasing mechanical integrity to the presomitic mesoderm, consistent with the tissue transitioning from a wetter to a dryer foam-like architecture. Our results show that cell-scale stresses fluctuate rapidly (within about 1 min), enabling cell rearrangements and effectively ‘melting’ the tissue at the growing end. Persistent (more than 0.5 h) stresses at supracellular scales, rather than cell-scale stresses, guide morphogenetic flows in fluid-like tissue regions. Unidirectional axis extension is sustained by the reported rigidification of the presomitic mesoderm, which mechanically supports posterior, fluid-like tissues during remodelling before their maturation. The spatiotemporal control of fluid-like and solid-like tissue states may represent a generic physical mechanism of embryonic morphogenesis.**

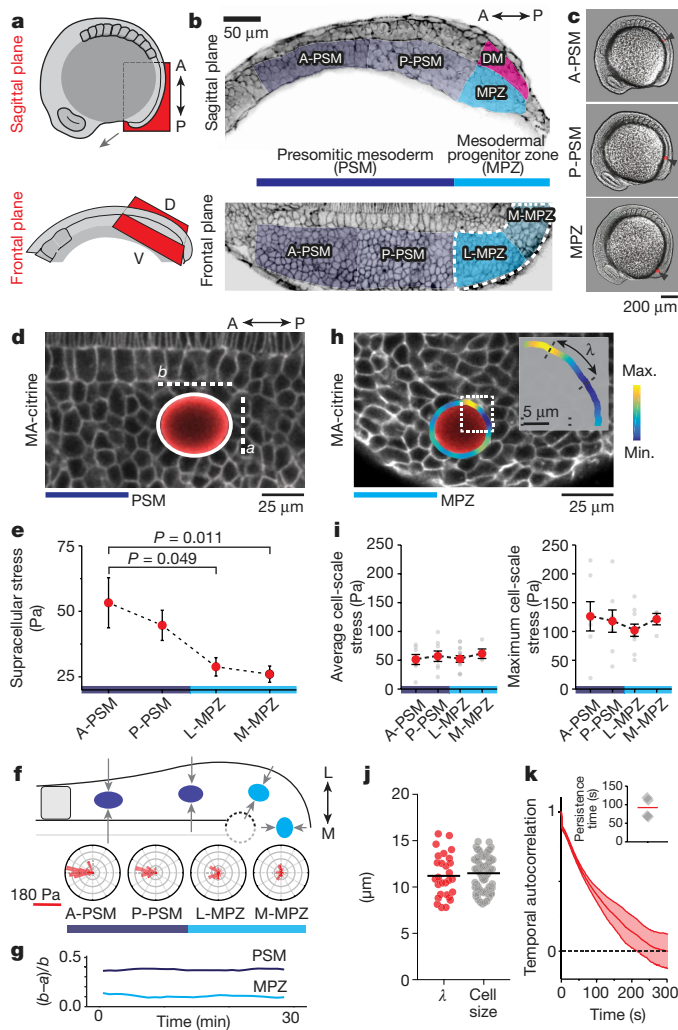
One of the hallmarks of animal development is the formation of the anteroposterior body axis, which occurs by nearly unidirectional elongation of tissues at the anterior and posterior body ends<sup>14</sup>. During posterior elongation, which does not depend on cell proliferation at early stages<sup>15</sup>, descendants of neuromesodermal progenitors move ventrally from the dorsal medial region to the paraxial mesodermal progenitor zone (MPZ) as mesodermal progenitors<sup>16,17</sup> (Fig. 1a, b). These progenitor cells progressively differentiate into mature mesodermal cells that are incorporated in the presomitic mesoderm (PSM) as body elongation proceeds. Both in amniotes and fish, cells display a graded reduction of their movements as they progress from the MPZ (referred to as posterior PSM in amniotes) to the PSM<sup>16,18</sup>, where they arrest and undergo a mesenchymal-to-epithelial transition that anticipates somite formation<sup>19</sup>. The observed patterns of cell motion in these kinematic studies<sup>16,18</sup> could potentially be caused by a gradient in cell or tissue mechanics along the anteroposterior axis. However, the physical mechanism causing such cellular movements and, more generally, sculpting the body axis, remains unknown.

Morphogenetic movements are generally believed to be guided by mechanical forces<sup>20</sup>. To establish the role of mechanical forces in vertebrate body elongation, we measured the endogenous mechanical stresses along the anteroposterior axis of zebrafish embryos using magnetically responsive oil microdroplets<sup>21</sup> (Fig. 1a–c; Methods). In the absence of an applied magnetic field, the deformations of droplets inserted between cells in the tissue provide a readout of the local mechanical stresses<sup>22</sup>. To detect potential spatial variations in mechanical stresses at supracellular scales, we analysed the ellipsoidal deformation of the droplets (Fig. 1d; Methods). Our analysis shows a posterior-to-anterior increase in the magnitude of supracellular stresses (Fig. 1e), with the axis of droplet elongation oriented mediolaterally in the medial MPZ (M-MPZ) and progressively reorienting along the anteroposterior axis in the PSM (Fig. 1f), consistent with the previously reported directions of morphogenetic flows<sup>16,18</sup>. Monitoring the ellipsoidal droplet deformation over time shows that supracellular stresses persist over more than 30 min (Fig. 1g), compatible with developmental timescales (around 0.5 to 1 h). These data indicate that supracellular stresses guide morphogenetic flows and reveal an increase in mediolateral constriction during PSM maturation, consistent with mediolateral thinning of the body axis<sup>16</sup> and anisotropy of nuclear deformation and cell shape (Extended Data Fig. 1).

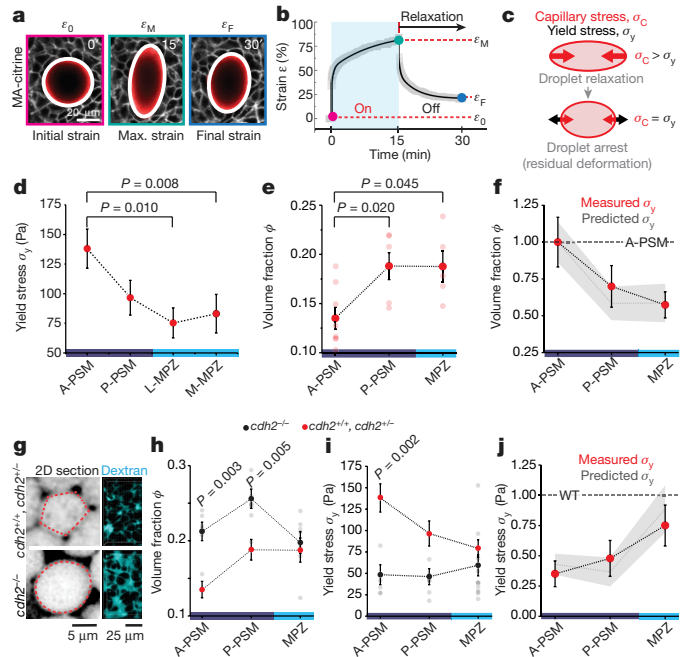
Beyond supracellular stresses, the deviations in the shapes of the droplets from ellipsoidal deformations display cell-sized stress inhomogeneities (Fig. 1h, j; Extended Data Fig. 2; Methods). Both the average and maximal values of these cell-scale stresses are uniform along the anteroposterior axis (Fig. 1i). Temporal autocorrelation analysis of these higher order droplet deformations shows that cell-scale stresses are short-lived, with a persistence time of approximately 1 min (Fig. 1k; Methods). These data show that cell-scale stresses are non-persistent at developmental timescales (0.5 to 1 h) and uniform throughout the tissue, suggesting that their role may be to introduce active stress fluctuations (causing cell ‘jiggling’; Supplementary Video 1), as previously proposed for 3D multicellular aggregates<sup>23</sup>.

Since morphogenetic flows also depend on the material properties of the tissue<sup>24,25</sup> and involve large tissue rearrangements, it is important to analyse the tissue mechanical response to large deformations. To do so, we applied a controlled, uniform magnetic field for a defined time period (15 min) to droplets previously inserted in the tissue<sup>21</sup> (Fig. 2a; Methods). Magnetic actuation caused large droplet deformations of more than one cell size, corresponding to applied strains in the 50–150% range (Fig. 2a, b). Upon removal of the magnetic field, droplets progressively relaxed towards a spherical shape due to the restoring force of interfacial tension (capillary stress,  $\sigma_c$ ), but arrested before reaching this state, displaying a residual deformation and revealing the presence of a yield stress  $\sigma_y$  in the tissue (Fig. 2a–c; Methods). The yield stress, which corresponds to the maximal mechanical stress that a material can sustain in a solid-like state before starting to flow<sup>3</sup>, provides a direct measure of the ability of the tissue to withstand sustained mechanical

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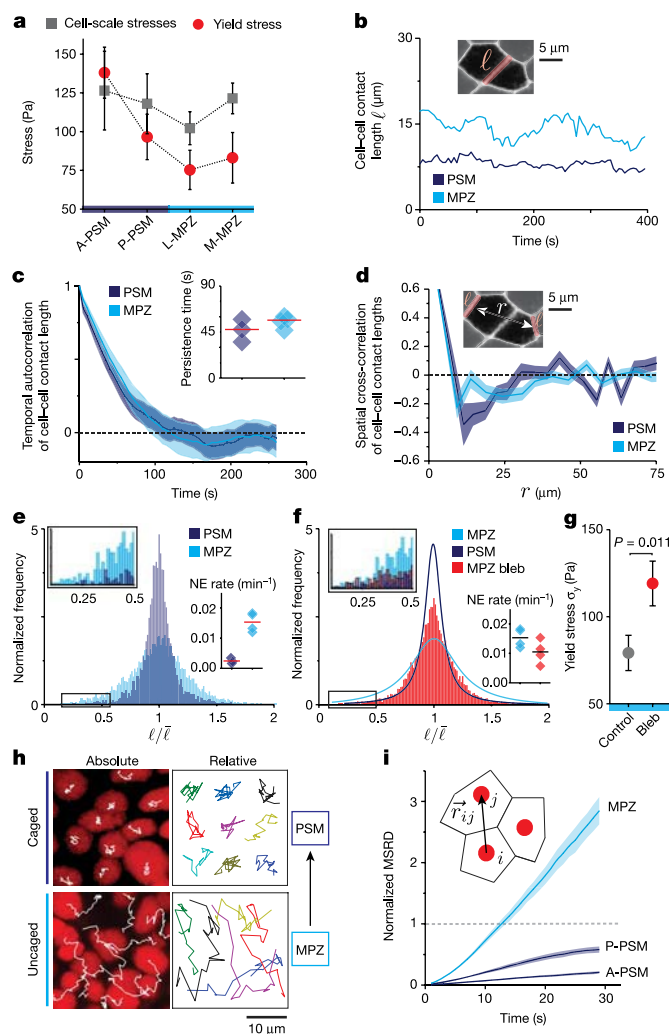
**Fig. 1 | Supracellular and cell-scale mechanical stresses during body axis elongation.** **a**, Sketch showing lateral views of a 10-somite stage embryo and sagittal and frontal anatomical planes. A, anterior; P, posterior; V, ventral; D, dorsal. **b**, Confocal sections along sagittal and frontal planes of posterior extending tissues in Tg(*actb2*:MA-citrine) embryos (inverted). The PSM is divided into anterior (A-PSM) and posterior (P-PSM) regions and the MPZ is divided into lateral (L-MPZ) and medial (M-MPZ) regions. The dorsal medial (DM) zone is dorsal to the MPZ. **c**, Embryos with droplets (red; arrows) located in the different regions. **d**, Elliptical fit (white;  $b$  and  $a$  being the long and short semi-axes) of a ferrofluid oil droplet (red) in the PSM of a Tg(*actb2*:MA-citrine) zebrafish embryo (no magnetic actuation). **e**, Magnitude of supracellular stresses along the anteroposterior axis ( $n = 9$  (A-PSM), 24 (P-PSM), 25 (L-MPZ), 27 (M-MPZ); mean  $\pm$  s.e.m.). Mann–Whitney  $U$ -test. **f**, Bottom, orientation of the long axis of the droplets with respect to the anteroposterior axis ( $n = 15$  (A-PSM), 12 (P-PSM), 11 (L-MPZ), 13 (M-MPZ)). M, medial; L, lateral. Top, sketch showing the average droplet orientations along the anteroposterior axis and the posterior-to-anterior increase in mediolateral constriction (arrows) in the PSM. **g**, Time evolution of the ellipsoidal droplet deformation,  $(b - a)/b$ . **h**, Ferrofluid droplet (red) in the MPZ of a Tg(*actb2*:MA-citrine) embryo. Inset, curvature values along the detected droplet contour (colour coded), with  $\lambda$  being the distance between consecutive curvature maxima and minima. **i**, Measured average (left) and maximal (right) cell-scale stresses along the anteroposterior axis ( $n = 7$  (A-PSM), 8 (P-PSM), 10 (L-MPZ), 4 (M-MPZ); mean  $\pm$  s.e.m.). **j**, Measured values of  $\lambda$  ( $n = 29$ ) and cell size ( $n = 100$  cells). Line indicates mean. **k**, Temporal autocorrelation of droplet shape deviations from the ellipsoidal mode ( $n = 2,062$  curvature time traces obtained from 4 embryos). Average half-life is approximately 1 min (inset;  $n = 4$ ; line indicates mean). Unless stated otherwise,  $n$  represents number of embryos.



**Fig. 2 | Mechanical integrity of the extending body axis.** **a**, Example of droplet dynamics during magnetic actuation. White lines indicate ellipse segmentation. Droplet actuation is characterized by an initial deformation or strain ( $\varepsilon_0$ ), a maximal strain, ( $\varepsilon_M$ ), and a final residual strain ( $\varepsilon_F$ ) (see Methods). **b**, Temporal evolution of the droplet strain during magnetic actuation. Experimental data points (grey) and fit (black line) are shown (Methods). **c**, The residual droplet deformation is set by the balance between the capillary stress ( $\sigma_C$ ) and the tissue yield stress ( $\sigma_Y$ ). **d**, **e**, Measured yield stress (**d**;  $n = 12$  (A-PSM), 12 (P-PSM), 13 (L-MPZ), 13 (M-MPZ)) and volume fraction of extracellular space ( $\phi$ ) (**e**;  $n = 8$  (A-PSM), 6 (P-PSM), 5 (MPZ)) along the anteroposterior axis. Mean  $\pm$  s.e.m.; Mann–Whitney  $U$ -test. **f**, Comparison between measured (red dots) and predicted (grey line and band representing mean  $\pm$  s.e.m.) yield stress along the anteroposterior axis, relative to the A-PSM. **g**, Confocal sections (inverted) and 3D reconstructions (Methods) of dextran-labelled extracellular space for control (*cdh2*<sup>+/+</sup>, *cdh2*<sup>+/-</sup>) and *cdh2*<sup>-/-</sup> embryos. **h**, **i**, Measured volume fraction of extracellular space (**h**;  $n = 6$  (A-PSM), 8 (P-PSM), 7 (MPZ)) and yield stress (**i**;  $n = 5$  (A-PSM), 5 (P-PSM), 12 (MPZ)) along the anteroposterior axis in *cdh2*<sup>-/-</sup> embryos compared to control embryos. Mean  $\pm$  s.e.m.; Mann–Whitney  $U$ -test. **j**, Measured (red dots) and predicted (grey line and band representing mean  $\pm$  s.e.m.) yield stress values in *cdh2*<sup>-/-</sup> embryos normalized to values for control embryos in each region. In all cases,  $n$  represents number of embryos.

loads, that is, its mechanical integrity. Performing these experiments along the body axis shows that the yield stress is spatially graded and minimal in the MPZ (Fig. 2d; Methods), revealing a posterior-to-anterior gradient of increasing tissue mechanical integrity that parallels PSM maturation.

The disordered cellular structure of tissues strongly resembles that of aqueous foams<sup>9,23,26</sup>, which display a finite yield stress when the volume fraction  $\phi$  of fluid between bubbles becomes lower than a critical value  $\phi_c$  at the jamming transition<sup>1,2</sup> (Supplementary Fig. 1). Following the analogy with foams, we explored the potential role of extracellular spaces in controlling tissue yield stress. We measured the volume fraction of extracellular spaces along the anteroposterior axis using fluorescent dextran (Methods) and observed decreasing spaces between cells away from the posterior end (Fig. 2e). Since cohesion between cells depends on their adhesion strength, and N-cadherin (also known as cadherin 2, encoded by the *cdh2* gene) has been shown to be essential for body axis elongation<sup>16,27</sup> and PSM maturation<sup>28</sup> in zebrafish, we measured the volume fraction of extracellular spaces and yield stress along the anteroposterior axis of *cdh2* loss-of-function mutants<sup>27</sup>. Disruption of N-cadherin-mediated adhesion resulted in larger extracellular spaces in the PSM, but not in the MPZ (Fig. 2g, h;



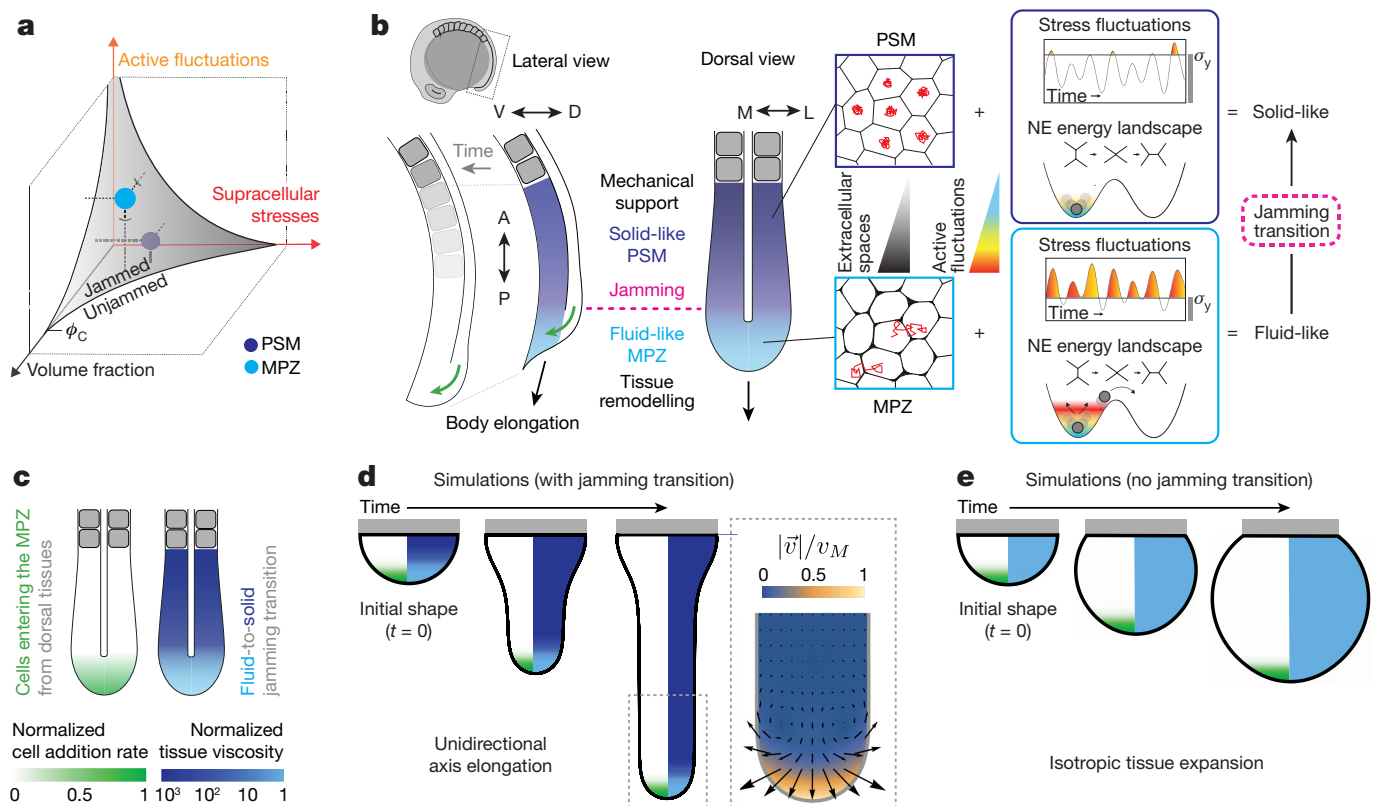
**Fig. 3 | Cell-cell contact length fluctuations and cellular movements along the anteroposterior axis.** **a**, Comparison of yield stress (Fig. 2d) and maximal cell-scale stresses (Fig. 1i) along the anteroposterior axis. **b**, Examples of time traces of cell-cell contact length  $\ell$  in the PSM and MPZ. **c**, **d**, Temporal autocorrelation of cell-cell contact length (**c**) and spatial cross-correlation of cell-cell contact lengths separated by a distance  $r$  (**d**) in the PSM and MPZ ( $n = 186$  (PSM), 54 (MPZ) cell-cell contacts). Correlation (persistence) timescale (**c**, inset; line indicates mean). **e**, Normalized frequency of cell-cell contact lengths in the PSM and MPZ ( $n = 6,969$  (PSM), 7,896 (MPZ) cell-cell contacts). Zoomed in distribution tail (box, left inset). Neighbour exchange (NE) rates (per cell) in PSM and MPZ (right inset;  $n = 14$  (PSM), 49 (MPZ) neighbour exchanges; line indicates mean). In **c–e**, cell-cell contacts were obtained from 3 and 4 embryos in PSM and MPZ, respectively. **f**, Normalized frequency of cell-cell contact lengths in the MPZ of embryos treated with 100  $\mu\text{M}$  blebbistatin (bleb) ( $n = 13,813$  cell-cell contacts from 4 embryos). Zoomed in distribution tail (box, left inset). Neighbour exchange rate in the MPZ region of blebbistatin-treated embryos compared to untreated embryos (right inset;  $n = 20$  neighbour exchanges from 3 embryos). **g**, Measured yield stress in the MPZ of embryos treated with blebbistatin (red;  $n = 12$  embryos) compared to untreated embryos (grey,  $n = 26$  embryos). Mean  $\pm$  s.e.m.; Mann–Whitney  $U$ -test. **h**, **i**, Tracks (**h**; absolute and relative; nuclear tracking) and normalized mean square relative displacement (MSRD) (**i**;  $n = 1,937$  (A-PSM), 1,776 (P-PSM) and 2,523 (MPZ) analysed cell pairs, from 5, 5 and 6 embryos, respectively) of cellular movements during a 30-min time window.

Extended Data Fig. 3). Accordingly, the yield stress in *cdh2* mutants is altered in the PSM but not in the MPZ (Fig. 2i). No significant spatial variation in yield stress was observed in *cdh2* mutants, indicating that N-cadherin-mediated adhesion is necessary to maintain the anteroposterior gradient in extracellular spaces and tissue mechanical integrity

(yield stress). In addition, the reported anteroposterior gradients in supracellular stresses (Fig. 1e) and nuclear anisotropy observed in control embryos are lost in *cdh2*<sup>−/−</sup> mutants (Extended Data Fig. 1), indicating that mediolateral constriction in the PSM is N-cadherin-dependent. Further supporting the analogy with foams, both the measured values of yield stress along the anteroposterior axis in wild-type embryos and the changes in yield stress between control and *cdh2*<sup>−/−</sup> mutant embryos can be accounted for (with no adjustable parameters) by assuming that such changes result solely from the observed variations in extracellular volume fraction in a tissue that behaves like a 3D-disordered monodispersed foam (Fig. 2f, j; Methods). Cell rearrangements induced by droplet actuation (Extended Data Fig. 4) further indicate that the yield stress arises from the jammed cellular environment, as in 3D disordered foams<sup>1</sup>. Whereas jamming transitions have recently been predicted by 2D vertex models of multicellular systems with no extracellular spaces<sup>10,11</sup> and observed in epithelial cell culture<sup>8</sup>, our measurements indicate that jamming in 3D embryonic tissues is consistent with classical jamming scenarios in foams, where the volume fraction of extracellular spaces has a prominent role<sup>1,2</sup>.

Unlike aqueous foams, however, living tissues are characterized by active cellular stresses that could locally unjam the system and fluidize the tissue, as observed in systems for which thermal fluctuations are relevant<sup>2,4,9</sup>. Comparison of the yield stress and the magnitude of cell-scale stresses shows that the maximal stress fluctuations are well above the yield stress in the MPZ, but below it in the anterior PSM (A-PSM, Fig. 3a), indicating that actively generated stress fluctuations may help to fluidize the tissue at its posterior end. Since neighbour exchanges are necessary to fluidize foam-like tissues<sup>1,9</sup> and actomyosin-driven fluctuations in cell-cell contact length cause cellular rearrangements<sup>20</sup>, we characterized the dynamics of cell-cell contacts both in the PSM and MPZ (Fig. 3b–f; Methods). Autocorrelation analysis shows that cell-cell contact lengths become temporally uncorrelated within less than 1 min and spatially uncorrelated beyond 2–3 cell sizes (Fig. 3c, d), consistent with our measurements of cell-scale stresses (Fig. 1i, k) and indicating that, at developmental timescales (approximately 0.5–1 h), cell-cell contact dynamics can be seen as short-lived, active fluctuations. As only large fluctuations can induce neighbour exchanges and fluidize the tissue, we measured the distribution of cell-cell contact lengths both in the PSM and MPZ tissues (Methods). Our results show a broader distribution in the MPZ, with large fluctuations being considerably more frequent in this region than in the PSM (Fig. 3e), both in control and in *cdh2* mutant embryos (Extended Data Fig. 5). Accordingly, we found that the neighbour exchange rate is 6 times higher in the MPZ than in the PSM (Fig. 3e, inset), consistent with the MPZ being in a fluid-like state and the PSM in a solid-like state. No systematic alignment of neighbour exchanges along a single spatial direction was observed (Extended Data Fig. 6), indicating that cell intercalation in these tissues does not contribute to posterior elongation. Interpreting active fluctuations as an effective temperature<sup>9,18,23</sup>, a key parameter in the control of jamming transitions<sup>2,4</sup>, it is possible to obtain the energy landscape of neighbour exchanges in both MPZ and PSM (Extended Data Fig. 7; Methods). In this framework, reducing fluctuations in cell-cell contact length (lower effective temperature) by impairing myosin II-dependent force generation at the cell cortex should render the MPZ more solid-like. Partial inhibition of myosin activity with blebbistatin led to smaller cell-cell contact length fluctuations (narrower distribution; Fig. 3f), decreased neighbour exchange rate (Fig. 3f, inset) and increased yield stress in the MPZ (Fig. 3g), thereby rigidifying the tissue and leading to a reduction in both cell movements and body elongation speed (Extended Data Fig. 8). These results indicate that while the PSM is jammed in a solid-like state, with weak cell-cell contact length fluctuations (low effective temperature) unable to fluidize the tissue, the stronger cell-cell contact length fluctuations (high effective temperature) in the MPZ cause substantial cell rearrangements that effectively ‘melt’ the tissue into a fluid-like state.

Since cellular movements result from the combined effect of active stresses and mechanical constraints (tissue material properties), their



**Fig. 4 | Physical mechanism of vertebrate body axis elongation.**

**a**, The MPZ (fluid-like) and PSM (solid-like) tissue states are represented in the jamming phase diagram<sup>2,4</sup>, which organizes jammed (solid-like) and unjammed (fluid-like) phases for varying volume fraction of extracellular spaces, supracellular stresses and active fluctuations (effective temperature). **b**, The higher cell–cell contact length fluctuations (high effective temperature) in the less constrained environment (more extracellular spaces) of the MPZ (light blue) drive cell rearrangements and cell mixing, effectively ‘melting’ the tissue in this region. As the paraxial mesoderm matures, the smaller extracellular spaces and low cell–cell contact fluctuations (low effective temperature) in the PSM (violet) rigidify the tissue via a jamming transition. Cells entering the MPZ from the dorsal medial region (green arrows; lateral view) cause the expansion of the fluid-like MPZ tissue, with the solid-like PSM acting as a rigid support that biases tissue expansion towards the posterior direction,

thereby elongating the body axis. Whereas persistent mediolateral supracellular stresses restrict lateral tissue expansion, the main role of non-persistent cell-scale stresses is to ‘melt’ the MPZ tissue, enabling its expansion and posterior elongation upon addition of new cells from the dorsal medial region. **c**, Sketch of the posterior body showing the input physical fields in the simulation. **d**, **e**, Time evolution of simulated tissue shapes (black outline) in the presence (**d**) and absence (**e**) of a jamming transition along the anteroposterior axis. The colour code in the right half of each shape corresponds to the spatial profile of the tissue viscosity (diverging in solid-like tissue regions), whereas the left half shows the anteroposterior profile of cell ingress rate into the MPZ from dorsal medial tissues. Grey rectangles represent a rigid boundary. Simulated morphogenetic flows (velocity field: direction, arrows; magnitude, colour coded) in the presence of a jamming transition leading to unidirectional body elongation (**d**, inset).

analysis informs about the mechanical state of the tissue<sup>6,8</sup>. Both direct observation (Fig. 3h) and analysis (Fig. 3i; Methods) of cellular movements in the tissue showed caged behaviour in the PSM, consistent with cells being trapped in a solid-like material, and uncaged cell mixing in the MPZ, consistent with this region being fluid-like. Our measurements indicate that the posterior-to-anterior transition between uncaged to caged cellular movements, consistent with previous observations in chicken<sup>18</sup>, results from an increase in physical constraints on cells, rather than a decrease in cellular stresses, as the spaces between cells are reduced anteriorly while maintaining a constant cell density (Extended Data Fig. 9; Methods).

Taken together, our direct *in vivo* measurements of tissue mechanics and analysis of cell rearrangements and movements are all consistent with the tissue behaving as a disordered, glassy material undergoing a jamming transition as the MPZ progressively rigidifies into the PSM (Fig. 4a, b). The solid-like state of the PSM helps to maintain tissue architecture and mechanically supports the extending, fluid-like end of the posterior body, as cells from the dorsal medial region are added to the MPZ and progressively elongate the body axis (Fig. 4b). Indeed, simulations of tissue morphogenesis based solely on first principles (Fig. 4c–e; Methods; Supplementary Note 1) show that unidirectional body axis elongation naturally occurs in the presence of the observed fluid-to-solid jamming transition (Fig. 4d; Supplementary Video 2).

In absence of jamming transition, the tissue expands isotropically like a growing spherical blob and no unidirectional axis elongation occurs (Fig. 4e; Supplementary Video 3). The predicted tissue morphogenetic flows in the presence of the reported jamming transition display high posterior-directed velocities at the posterior end, no tissue flow in the A-PSM, and the existence of two counter-rotating vortices as the tissue transits from fluid-like to solid-like states (Fig. 4d, inset). Remarkably, all these predicted features in tissue flows have been observed experimentally<sup>16</sup>, indicating that the fluid-to-solid jamming transition is essential for proper unidirectional axis elongation. Phenotypes such as curved tails and the reduced elongation speed in *cdh2* mutants (Extended Data Fig. 8) can be explained by the reported loss of mechanical integrity in the PSM (Fig. 2i) during body axis elongation.

More generally, the spatiotemporal control of fluid-like and solid-like tissue states, enabling or restricting morphogenetic flows, represents a novel mechanism of tissue morphogenesis. The previously observed oscillations in actomyosin contractility during apical constriction<sup>29</sup> in *Drosophila* epithelia may be necessary to overcome a yield stress and fluidize an otherwise jammed tissue<sup>13</sup>. Beyond cellular jamming, transitions between fluid-like and solid-like states in tissues containing extracellular matrix between cells are likely to occur via control of the physical-chemical state of the extracellular matrix. As suggested by D’Arcy Thompson a century ago<sup>26</sup>, the shaping of living tissues shares

fundamental physical principles with the sculpting of many inert materials, in which fluid-to-solid transitions are necessary to mould them into functional shapes.

## Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of data availability and associated accession codes are available at <https://doi.org/10.1038/s41586-018-0479-2>.

Received: 8 September 2017; Accepted: 3 August 2018;

Published online 5 September 2018.

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**Acknowledgements** We thank E. Sletten for sharing custom-made fluorinated dyes. We also thank all laboratory members and the UCSB Animal Research Center for support. P.R. thanks B. Aigouy for assistance with Tissue Analyzer. A.M. thanks EMBO (EMBO ALTF 509-2013), Errett Fisher Foundation and Otis Williams Fund for financial support. This work was partially supported by the National Science Foundation (CMMI-1562910) and the Eunice Kennedy Shriver National Institute of Child Health and Human Development of the National Institutes of Health (R21HD084285; R01HD095797).

**Reviewer information** Nature thanks J. Fredberg, P.-F. Lenne, O. Pourquié and the other anonymous reviewer(s) for their contribution to the peer review of this work.

**Author contributions** A.M. and O.C. designed research; A.M., H.J.G., D.A.K. and F.S. performed experiments; A.M., H.J.G., P.R., E.S. and J.G. analysed the data; P.R. and O.C. performed theoretical interpretation of experiments; E.K.C. and P.R. performed simulations; A.A.L. assisted with droplet generation; A.M. and O.C. wrote the paper; O.C. supervised the project.

**Competing interests** The authors declare no competing interests.

## Additional information

**Extended data** is available for this paper at <https://doi.org/10.1038/s41586-018-0479-2>.

**Supplementary information** is available for this paper at <https://doi.org/10.1038/s41586-018-0479-2>.

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## METHODS

**Zebrafish husbandry, fish lines, and experimental manipulations.** Zebrafish (*Danio rerio*) were maintained as previously described<sup>30</sup>. Animals were raised and experiments were performed following all ethical regulations and according to protocols approved by the Institutional Animal Care and Use Committee (IACUC) at the University of California, Santa Barbara. For ubiquitous labelling of cell membranes, we used either Tg(*actb2:MA-citrine*) embryos<sup>27,31</sup> or embryos that had been injected with membrane-GFP mRNA at the one-cell stage. For experiments to perturb the amount of extracellular spaces we used *cdh2<sup>tm101/tm101</sup>* embryos (also known as parachute (*pac*) mutants<sup>27</sup>), which feature a truncated N-cadherin (*Cdh2*) extracellular domain and have been shown to display cell–cell adhesion defects<sup>27</sup>. For tracking of cell movements, one-cell stage embryos were injected with H2B-RFP mRNA.

**Generation and injection of ferrofluid droplets.** The composition and preparation of ferrofluid droplets was identical to those previously described<sup>21</sup>. In brief, we prepared biocompatible fluorocarbon-based ferrofluids by diluting DFF1 (Ferrotec) in Novec 7300 (3M) at varying concentrations to achieve the necessary magnetic stresses to generate the desired droplet deformations (applied strains). To prevent non-specific adhesion between the cells and the ferrofluid droplets, a fluorinated Krytox-PEG(600) surfactant (008-fluorosurfactant, RAN Biotechnologies<sup>32</sup>) was diluted in the ferrofluid at a 2.5% (w/w) concentration. The ferrofluid used in each experiment was calibrated as previously described<sup>21</sup>, so that the stresses applied by the ferrofluid droplet on the surrounding material in the presence of a uniform magnetic field were quantitatively known. Ferrofluid droplets were generated inside embryos by direct injection of the ferrofluid in the tissue of interest, as previously described<sup>21</sup>. Control of droplet size was achieved by controlling the injection pressure and time of the injection pulse. Droplets were injected in the MPZ tissue at the 4- and 6-somite stages for measurements in the PSM and MPZ, respectively. Droplets were imaged at least 2 h after injection to allow enough time for the tissue to recover from the injection. The mechanical relaxation timescales in the MPZ and PSM are short (~1 min; Supplementary Fig. 2b) and cell rearrangements are observed in timescales shorter than 10 min (Fig. 3e, inset). Therefore, our experiments, performed 2 h after injection, allow enough time for the tissue to remodel. Indeed, no trace of the injection is observed in the tissue when imaged. Importantly, we have previously shown that injection and actuation of magnetic droplets located in the MPZ and PSM tissues do not affect normal developmental processes and, especially, body axis elongation<sup>21</sup>.

**Imaging.** Embryos were mounted for imaging in 0.8% low-melting agarose and imaged at 25 °C using a laser scanning confocal (LSM 710, Carl Zeiss Inc.). Images were taken at 2.5-s intervals for droplet actuation experiments and at 60-s intervals for analysis of cell movements, using a 40× water immersion objective (LD C-Apochromat 1.1 W, Carl Zeiss) or a 10× air objective (EC Epiplan-Neofluar 0.25, Carl Zeiss). Imaging of ferrofluid droplets in the embryo was done as previously reported<sup>21</sup>. Ferrofluid droplets were fluorescently labelled using a custom-synthesized fluorinated rhodamine dye<sup>33</sup>, which was dissolved in the fluorocarbon-based ferrofluid oil at a final concentration of 37 μM.

**Measurement of supracellular (tissue-level) stresses.** To quantify supracellular stresses, we measured the droplet ellipsoidal deformation, as this deformation mode reveals mechanical stresses that occur at the length scale of the droplet diameter (and slightly larger scales), which we purposely made larger than the cell size (Fig. 1d; droplet diameter: 44 ± 6 μm; cell diameter: 11 ± 5 μm; mean ± s.e.m.). We used ferrofluid droplets because their interfacial tension can be calibrated *in vivo*<sup>21</sup>. No magnetic field was applied on the droplet before or during the measurement of supracellular stresses. Moreover, we prevented cell adhesion to the droplets to reveal the stresses associated solely with (supracellular) morphogenetic flows, mirroring previous studies in inert fluids<sup>34</sup>. Two hours after injection, the droplet mid-plane was imaged using confocal microscopy (Zeiss LSM 710, Zeiss). To quantify the ellipsoidal deformation of the droplet (and neglect high order deformations), we fitted an ellipse to the measured droplet shape, as previously described<sup>21</sup>. The value of anisotropic stresses for the ellipsoidal droplet deformation is given by  $\sigma_T^A = 2\gamma(H_b - H_a)$ , where  $H_b$  and  $H_a$  are the mean curvatures of the droplet at the intersection of the two principal axis with the ellipsoid and  $\gamma$  is the droplet's interfacial tension (Fig. 1d), as previously established<sup>22,35</sup>. The mean curvatures  $H_b$  and  $H_a$  are given by  $H_b = b/a^2$  and  $H_a = 1/2a + a/(2b^2)$ , with  $b$  and  $a$  being the long and short semi-axis of the fitted ellipse, respectively (Fig. 1d). At the end of each experiment, that is, after recording the shape of each droplet, we measured its interfacial tension  $\gamma$  *in situ*, within the developing embryo, as previously described<sup>21</sup>.

To obtain the direction of droplet deformation, we measured the angle between the direction defined by the long axis of elliptical droplet deformation and the anteroposterior axis of the embryo using ImageJ.

While droplets cannot measure isotropic stresses because of droplet incompressibility<sup>22</sup>, the ellipsoidal droplet deformation provides a quantitative measure of the difference in stresses along the direction of droplet elongation and the

perpendicular direction in the observation plane (the principal directions of the deformation). Ellipsoidal droplet deformations can, in general, be caused both by shear stresses or by stress differences along principal axes of droplet deformation (stress anisotropy).

**Droplet shape segmentation and measurement of in-plane curvature.** Confocal sections through the middle of a ferrofluid droplet were obtained by confocal microscopy, as described above. We then used a custom-made MATLAB code (adapted from a previously published code<sup>35</sup>) to obtain the coordinates of the droplet contour (segmentation) and measure the in-plane curvature  $\kappa(s)$  along the droplet contour, with  $s$  being the arc length along the contour. We first applied a Gaussian low-pass filter on the original raw image. The drop shape was identified from the filtered image using active contour segmentation to generate a mask. The edge was located by convolving the mask with the Sobel operator. The locations of edge pixels were then converted to ordered polar coordinates, which were smoothed using a moving average filter with a span of 5 pixels. We resampled the coordinates at even spacing using shape-preserving piecewise cubic interpolation. The curvature  $\kappa(s)$  along the droplet contour was obtained by cubic fitting of edge coordinates over a small neighbourhood of each point along the contour.

**Measurement of stresses associated with deviations from the ellipsoidal droplet deformations.** To measure the value of stresses associated with deformation modes of higher order than the ellipsoidal deformation mode (droplet deformations at length scales smaller than the droplet size), we used a similar procedure to the one described<sup>22</sup> to obtain cellular stresses from 2D confocal sections. Maximal and minimal values of the curvature along the droplet contour were spaced on average by the measured cell size (Fig. 1h, j), confirming that these deformations are associated with spatial variations in stresses occurring at the cell scale. In brief, we first segmented the 2D droplet shape and obtained the in-plane curvature  $\kappa(s)$  along the droplet contour, as described above. We then calculated the deviations  $\delta\kappa(s) \equiv \kappa(s) - \kappa_e(s)$  of the curvature along the contour from the curvature  $\kappa_e$  of the elliptical deformation mode. We fitted an ellipse to the contour coordinates using the MATLAB function `EllipseDirectFit` (written by N. Chernov based on a previous algorithm) to determine the elliptical curvature  $\kappa_e$ , and then calculated directly  $\delta\kappa(s)$  along the droplet contour. We then detected the maxima and minima of  $\delta\kappa$  along the contour and defined the amplitude of curvature deviations from the elliptical mode as the difference between a consecutive maximum and minimum of curvature,  $\delta\kappa^{\max}$  and  $\delta\kappa^{\min}$ , respectively, along the droplet contour, namely  $\Delta\kappa_C \equiv \delta\kappa^{\max} - \delta\kappa^{\min}$ . We also defined the maximum amplitude of curvature deviations from the elliptical deformation mode,  $\Delta\kappa_C^{\max} \equiv \delta\kappa^{\text{absmax}} - \delta\kappa^{\text{absmin}}$ , where  $\delta\kappa^{\text{absmax}}$  and  $\delta\kappa^{\text{absmin}}$  correspond to the absolute maximum and absolute minimum in  $\delta\kappa$  along the droplet contour. The average and maximal values of the stresses associated to cell-sized deviations from the ellipsoidal mode,  $\sigma_C^{\text{average}}$  and  $\sigma_C^{\max}$  respectively, are given by  $\sigma_C^{\text{average}} = \langle 2\gamma\Delta\kappa_C \rangle$  (average, indicated by the angled brackets, was calculated over multiple consecutive maxima and minima for a single droplet and also over multiple droplets in the same region of the tissue in different embryos) and  $\sigma_C^{\max} = \langle 2\gamma\Delta\kappa_C^{\max} \rangle$  (average was done over multiple droplets in the same region of the tissue in different embryos). As previously noted<sup>22</sup>, this calculation assumes no major structural anisotropies in the tissue.

To measure the length scale associated with shape deviations from the elliptical deformation mode, we obtained the locations of maxima and minima of  $\delta\kappa(s)$  along a droplet's contour, and calculated the contour distance  $\lambda$  between consecutive maxima and minima (Fig. 1h).

**Measurement of persistence in supra-cellular (tissue-level) stresses.** To quantify the persistence timescale of tissue-level stresses, we imaged equatorial sections of magnetic droplets (no applied magnetic fields, as described above) at time intervals of one second for up to 30 min. We fitted ellipses to droplet sections at each time point and reported the droplet aspect ratio as a function of time. We observed no substantial change in the average droplet aspect ratio over the course of 30 min, indicating that tissue-level stress anisotropies persist in the tissue for longer than at least 30 min.

**Measurement of persistence in cell-scale stresses.** To obtain the persistence of cell-scale stresses, we performed time-lapses of ferrofluid droplets (no magnetic actuation) inserted in the MPZ tissue at 2.5-s time intervals, for 10 to 16 min. We segmented those droplets at each time point and obtained the in-plane curvature  $\kappa(s, t)$  along the contour (parameterized with the contour length  $s$ ), as well as the curvature deviations from the elliptical droplet deformation,  $\delta\kappa(s, t)$ , as described above. For each time-lapse, we resampled  $\delta\kappa(s, t)$  at equal angular ( $\theta$ ) spacing, to obtain  $\delta\kappa(\theta, t)$ . We then calculated the temporal autocorrelation  $C_{\delta\kappa}(\tau)$ , namely

$$C_{\delta\kappa}(\tau) = \frac{\langle (\delta\kappa(\theta, t + \tau) - \langle \delta\kappa(\theta, t + \tau) \rangle_{\theta, t}) (\delta\kappa(\theta, t) - \langle \delta\kappa(\theta, t) \rangle_{\theta, t}) \rangle_{\theta, t}}{\sqrt{\langle (\delta\kappa(\theta, t + \tau) - \langle \delta\kappa(\theta, t + \tau) \rangle_{\theta, t})^2 \rangle_{\theta, t}} \sqrt{\langle (\delta\kappa(\theta, t) - \langle \delta\kappa(\theta, t) \rangle_{\theta, t})^2 \rangle_{\theta, t}}}$$

We reported the value of the timescale  $\tau_{1/2}$  for which  $C_{\delta\kappa}(\tau)$  drops below 0.5 (or  $C_{\delta\kappa}(\tau_{1/2}) = 0.5$ ), which corresponds to its half-life and provides a measure of the timescale of correlation loss or, equivalently, the persistence timescale of cell-scale droplet deformations.

**Measurement of average cell size.** Cell size was calculated using the polygonal drawing tool of ImageJ to outline the cells of each region and calculate the area of each selection. The average cell diameter was obtained by the diameter of a circle with the same area as that measured from the polygonal cell shape. The average cell size in a given region of the tissue was obtained from the ensemble average of 100 cells (approximately 35 in each region, A-PSM, P-PSM and MPZ).

**Measurement of cell shape and nuclear anisotropy.** Maximum intensity projections of 3 consecutive planes of a confocal stack, spanning in total  $6\ \mu\text{m}$  in  $z$ , were subjected to thresholding and converted to binary images using ImageJ. Subsequently, the ImageJ plug-in 'Fill Holes' was applied and ellipses were fitted to the H2B-labelled nuclei. Nuclear anisotropy was given by the aspect ratio of the fitted ellipse. To quantify the orientation of nuclear elongation, we used ImageJ to measure the angle between the axis of nuclear elongation (given by the long axis of the fitted ellipse) and the anteroposterior axis. The direction of the anteroposterior axis was revealed by the notochord in the images. To obtain cell shape anisotropy, we first measured the length of cell–cell contacts oriented within  $18^\circ$  degrees of the anteroposterior and mediolateral axes. To do so, we used Tissue Analyzer (see 'Spatial and temporal autocorrelation of cell–cell contact length') to obtain the cell–cell contact lengths versus their orientation in 2D confocal sections of different regions.

**Magnetic actuation of ferrofluid microdroplets.** Actuation of ferrofluid droplets was done with the same equipment and methods as described<sup>21</sup>. In brief, we deformed a ferrofluid droplet previously injected in the tissue using an applied, uniform and constant magnetic field (as previously described<sup>21</sup>), thereby generating a controlled local strain in the tissue. We applied a uniform and constant magnetic field for 15 min, as this time is considerably longer than all measured stress relaxation timescales in the tissue<sup>21</sup> (Supplementary Fig. 2b). Ferrofluid droplets deform into ellipsoids when a uniform, constant magnetic field is applied<sup>21,36</sup>. We monitored the time evolution of the droplet deformation and measured the initial ( $b_0$ ), maximal ( $b_M$ ) and final ( $b_F$ ) droplet semi-axis in the direction of the applied magnetic field (Fig. 2a, b). The final droplet semi-axis ( $b_F$ ) was determined from the asymptotic deformation of the droplet after relaxation following the removal of the applied magnetic field. We defined the droplet strain as  $\varepsilon \equiv (b - R)/R$ . With this definition, the initial strain  $\varepsilon_0$ , the maximum applied strain  $\varepsilon_M$  and the final strain  $\varepsilon_F$  are given by  $\varepsilon_0 \equiv (b_0 - R)/R$ ,  $\varepsilon_M \equiv (b_M - R)/R$  and  $\varepsilon_F \equiv (b_F - R)/R$ , respectively. We applied maximal mechanical strains in the 50–150% range, depending on the droplet's interfacial tension and the magnitude of the applied magnetic field.

**Measurement of local yield stress with ferrofluid microdroplets.** To measure yield stress, we applied strong magnetic fields to generate ellipsoidal droplet deformations larger than at least one cell size, as in these conditions the tissue flows irreversibly. Upon removal of the magnetic field, the capillary stress of the droplet pulls the droplet back towards the undeformed, spherical droplet shape (Fig. 2c). The capillary stress,  $\sigma_c$ , is the mechanical normal stress that tries to restore the spherical droplet shape because of the presence of an interfacial tension and depends on how deformed the droplet is: the larger the droplet deformation, the larger the capillary stress, as its value at a given point of the droplet's surface is proportional to the mean curvature  $H$  of the droplet at that point. Starting from the maximal droplet deformation induced by the applied magnetic field (maximal strain,  $\varepsilon_M$ ), the capillary stresses driving droplet relaxation are large and become smaller as the droplet shape relaxes towards the sphere. In the absence of a yield stress in the tissue, the droplet would relax to the spherical shape. However, the presence of a yield stress halts the relaxation of the droplet at the aspect ratio for which the capillary stress is equal to the yield stress (Fig. 2c), preventing further droplet relaxation. At this point, the yield stress balances the droplet's capillary stresses (Fig. 2c). Since the interfacial tension of the droplet is measured in situ, the final aspect ratio of the arrested droplet serves as a direct readout of the yield stress. Importantly, we applied the magnetic field in the perpendicular direction to the droplet pre-deformation, so that the final aspect ratio in the direction of the magnetic field could not be attributed to the supracellular tissue anisotropy in mechanical stress that deformed the droplet initially. Moreover, to allow accurate comparisons between the measured anisotropic stresses and yield stress measurements, we defined the anisotropic yield stress as the anisotropic stress that results from the residual droplet deformation that remains after droplet relaxation is halted by the yield stress (Fig. 2c). The anisotropic yield stress is given by  $\sigma_y = 2\gamma(H_{b,r} - H_{a,r})$ , where  $H_{b,r}$  and  $H_{a,r}$  are the mean curvatures of the residual droplet shape (deformation) at the intersection of the principal axis with the ellipsoid. Since the residual droplet shape is a prolate spheroid with long and short semi-axes  $b_r$  and  $a_r$ , respectively,  $H_{b,r}$  and  $H_{a,r}$  read  $H_{b,r} = b_r/a_r^2$  and  $H_{a,r} = 1/2a_r + a_r/(2b_r^2)$ . The interfacial tension of each droplet was measured directly in situ, within the developing embryo, as previously described<sup>21</sup>. Using the measured value of

the interfacial tension for each droplet and the mean curvatures of the residual droplet deformation, we obtained the value of the yield stress,  $\sigma_y$ .

The measured values of the yield stress did not depend on the extent of droplet deformation before starting the relaxation process (maximal droplet deformation or strain,  $\varepsilon_M$ ; Fig. 2a, b), ruling out potential memory effects that could occur in hysteretic plastic materials (Extended Data Fig. 10).

**Measurement of volume fraction of extracellular space.** To obtain the volume fraction of the extracellular space, we injected dextran–Alexa Fluor 488 (10,000 MW) in the MPZ of 9-somite stage embryos. After 30–45 min embryos were mounted and imaged. 3D reconstructions of different regions of the paraxial mesoderm were analysed in Imaris (Bitplane), using the surface reconstruction algorithm. The volumes encapsulated by the Alexa Fluor 488 segmented surfaces were then divided by the total volume analysed. Injection of dextran–Alexa Fluor 488 in the PSM of 9-somite stage embryos led to consistent results. The extracellular spaces are also visible by fluorescently labelling cell membranes (Extended Data Fig. 4), but the quantification of extracellular spaces was considerably more difficult and less accurate in this case.

**Theoretical predictions of yield stress from volume fraction of extracellular space.**

To obtain the predicted values of yield stress (gradient in wild type, Fig. 2f, and change between *cdh2* mutants and controls, Fig. 2j) from the measured values of the volume fraction  $\phi$  of extracellular space, we assumed the tissue to behave as a 3D disordered monodispersed foam<sup>1</sup> for which the yield stress is given by  $\sigma_y = \sigma_y^0(\phi_C - \phi)^2$ , where  $\phi_C$  is the critical volume fraction ( $\phi_C = 0.36$  for 3D disordered monodispersed foams<sup>1</sup>) and  $\sigma_y^0$  is a stress scale that depends on the interfacial tension of bubbles and their size. We assumed the magnitude of cell–cell contact tensions (playing the role of the bubble interfacial tension in an aqueous foam) to be constant along the anteroposterior axis, as our measurements indicated that cell-scale stresses are largely uniform in the tissue. The cell size was also assumed to be constant, because our measurements indicate no significant changes along the anteroposterior axis (Extended Data Fig. 9). The comparison between the experimental values of the yield stress and the predicted ones has no adjustable parameters (Fig. 2f, j).

**Measurement and analysis of 3D cellular movements.** To track cellular movements in 3D, both in control and *cdh2* mutant embryos, we first fluorescently labelled cellular nuclei by injecting H2B-mRFP mRNA into one-cell stage embryos. At the 10-somite stage, embryos were manually dechorionated and mounted in 0.8% low melting point agarose in a 35-mm glass-bottom dish (MatTek Corporation). Each region (A-PSM, P-PSM and MPZ) was imaged at 25 °C using a 40× water immersion objective (LD C-Apochromat 1.1W, Carl Zeiss Inc.), with  $z$ -stacks acquired at 1-min intervals for 30 min total.  $z$ -stacks were cropped to include only cells corresponding to the region of interest. Loss of signal intensity over time was corrected with the 'EMBL Bleach Correction' plugin for ImageJ using the histogram matching method. The 3D nuclei positions  $r_i$  (with  $i$  specifying the nuclei) at each time point were determined using the Imaris (Bitplane) automated spot detection. We then used the Imaris (Bitplane) tracking algorithm to obtain the trajectories of the nuclei,  $r_i(t)$ . For subsequent analysis, the results were filtered by track duration to include only those tracks that were maintained over the full course of the time lapse. We then used the nuclei 3D trajectories  $r_i(t)$  as a proxy for the trajectories of the cells. To eliminate the effect of translational and rotational motions of the whole tissue/embryo, we analysed the relative cell distances  $r_{ij} \equiv r_i - r_j$ . We evaluated the mean-square displacement of the relative distances, or mean-square relative displacements (MSRD), from their initial values, that is,  $\text{MSRD}(t) \equiv (r_{ij}(t) - r_{ij}(0))^2$ , for  $(i, j)$  pairs, which were nearest neighbours at  $t = 0$ . We evaluated the normalized MSRD( $t$ ), namely  $(\text{MSRD}(t) - \text{MSRD}(t = 0))/d^2$  (with  $d$  being the average cell size), for every  $(i, j)$  nearest pair in each region (A-PSM, P-PSM, and MPZ) of a given embryo, and averaged the results over all the pairs in the same region for each embryo.

**Spatial and temporal autocorrelation of cell–cell contact length.** To determine the characteristic timescale of changes in cell–cell contact lengths, we acquired a single confocal section of embryos injected with memGFP mRNA for 30 min with time resolution of 5 s. We detected the location of cell vertices and cell–cell contact lengths in the images using Tissue Analyzer<sup>37</sup> (formerly known as Packing Analyzer), a Fiji plugin capable of segmenting 2D tissue images and quantifying cell–cell contact lengths and position of vertices over time. For each embryo, we segmented a region of interest (ROI) in the PSM or tailbud (MPZ) for which cell–cell contacts were trackable over a 400 s time period. We then used the Tissue Analyzer package to obtain the contour length of cell–cell contacts and the  $(x, y)$  positions of the vertices. Using the timeseries of the contour length of cell–cell contacts and the  $(x, y)$  positions of the vertices, we calculated the temporal autocorrelation function, namely

$$C_\ell(\tau) \equiv \frac{\langle (\ell(t+\tau) - \langle \ell(t+\tau) \rangle)_i (\ell(t) - \langle \ell(t) \rangle)_i \rangle_t}{\sqrt{\langle (\ell(t+\tau) - \langle \ell(t+\tau) \rangle)_i^2 \rangle_t} \sqrt{\langle (\ell(t) - \langle \ell(t) \rangle)_i^2 \rangle_t}}$$

where  $\ell$  is the cell–cell contact length. To reduce numerical errors, each average  $\langle \bullet \rangle_t$  was evaluated with the data in a time interval  $(0, T - \tau)$ , with  $T$  being the duration of the experiment, and  $\langle \bullet \rangle_{t+\tau}$  was evaluated with the data in the time interval  $(\tau, T)$ . We analysed each ROI and obtained the correlation function for each embryo separately. The obtained autocorrelation functions were nearly exponential in all cases (Fig. 3c). We therefore obtained the characteristic timescale of the decay of the correlation function by fitting an exponential function. The reported characteristic timescale was obtained from a weighted average of the timescales measured in different embryos, with the weights being the inverse of the variance. To obtain the spatial cross-correlation, a similar analysis was done, but correlating instead every two pairs of cell–cell contact lengths in the region of the analysed tissue. The distance between cell–cell contact pairs was given by the norm of the vector connecting the midpoints of the two cell–cell contacts (Fig. 3d).

**Normalized frequency (distribution) of cell–cell contact length fluctuations and energy landscape of neighbour exchanges.** We obtained the distribution of normalized cell–cell contact lengths, and the corresponding dimensionless energy landscape, which characterizes neighbour exchanges. For the ROIs described above (in ‘Spatial and temporal autocorrelation of cell–cell contact length’), we selected all cell–cell contacts that persist for at least 100 s, and obtained their time-dependent normalized lengths  $\ell/\bar{\ell}$ , where  $\ell$  is the cell–cell contact length at a given time point and  $\bar{\ell}$  is the average length of each individual cell–cell contact over time. We then combined all the values of  $\ell/\bar{\ell}$  for all times and cell–cell contacts into a single, normalized frequency distribution  $p(\ell/\bar{\ell})$ , which peaks around 1 (Fig. 3e). Assuming that the cell–cell contact length fluctuations are probing an energy landscape that varies much more slowly (equilibrium approximation) before they undergo neighbour exchanges, we obtained the dimensionless energy landscape of neighbour exchanges by calculating  $-\ln[p(\ell/\bar{\ell})]$  (Extended Data Fig. 7). Since the statistics of cell–cell contact length fluctuations is small for values close to 0 (that is, close to a neighbour exchange event), this method provides the dimensionless energy landscape close to the bottom of the energy landscape but does not allow to obtain the size of the energy barrier.

**Measurement of neighbour exchange rates and orientation.** To quantify the neighbour exchange rate in the tissue, we analysed 2D confocal sections of untreated embryos and blebbistatin-treated embryos, both in the PSM and MPZ. We counted the number of cell–cell contacts that underwent a neighbour exchange in the plane of observation. We obtained the neighbour exchange rate per cell by normalizing the number of measured neighbour exchanges in the observed region with the average cell area and observation time (30 min). To obtain their orientation, we measured the angle between the cell–cell contact and the direction of the anteroposterior axis before neighbour exchange. The direction of the anteroposterior axis was revealed by the notochord in the images.

**Inhibition of myosin activity.** Myosin activity was inhibited by adding in the embryo medium blebbistatin (Tocris) at the final concentration of 100  $\mu\text{M}$ . Treatments were started at the four-somite stage.

**Measurement of cell density.** To measure cell number density (number of cells per unit volume), we performed confocal 3D scans of the different regions (MPZ, P-PSM and A-PSM) in embryos with fluorescently labelled nuclei (same procedure as described above in ‘Measurement and analysis of 3D cellular movements’). We then used Imaris software to detect the location of the nuclei in the tissue (approximately 300 cells per imaged volume). After obtaining the 3D coordinates of each nucleus in a volume, we defined a box of smaller size than the full scanned volume and measured the number of nuclei in the box. Moving the small box inside the

originally scanned volume, we obtained statistics for the density measurement. This procedure was repeated with varying box sizes to ensure that the measured value of the density did not depend on the box size.

**Actin staining.** Ten-somite stage embryos were dechorionated and fixed in 4% paraformaldehyde for 1.5 h at room temperature. Embryos were then washed 2–3 times with PBS. One drop of ActinRed was added to the PBS solution (500  $\mu\text{l}$ ) containing the embryos. After 30 min, the embryos were washed with new PBS 3 times and prepared for imaging as described above.

**Simulations.** We simulated the physical expansion of a 2D dorsal–ventral projection of the PSM and MPZ tissues at supracellular length scales and developmental time scales using finite element methods, accounting for the fluid-to-solid jamming transition (or its absence, depending on the simulation) and the entrance of cells to the MPZ from dorsal medial tissues (see Supplementary Note 1 for details). The fluid-to-solid jamming transition was simulated by a sharp increase in the tissue viscosity along the anteroposterior axis, with a diverging (very large) viscosity simulating the solid-like tissue state. The dynamics of the tissue is governed by fundamental physical laws, namely momentum conservation (which reduces here to local force balance because inertial terms are negligible for embryonic tissues) and mass balance. We solved the equations using COMSOL Multiphysics 5.3 software, starting from a semi-circular tissue shape and with a fixed rigid boundary at the most anterior end. The solutions provided the time evolution of the tissue shape and velocity field.

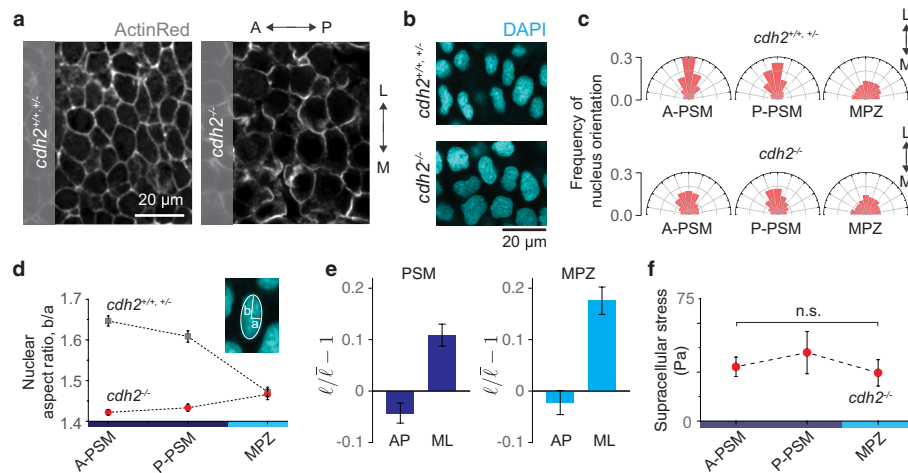
**Statistics.** In experiments involving zebrafish embryos, the sample size was chosen so that new data points would not significantly change the standard deviation. No samples were excluded from the analysis and the analysis of all the data was done by automated software to ensure full blinding and avoid biases in the analysis. No randomization of the data was used.

**Reporting summary.** Further information on research design is available in the Nature Research Reporting Summary linked to this paper.

**Code availability.** A previously published version of the custom-made MATLAB code to analyse droplet deformations<sup>33</sup> is available on GitHub at <https://github.com/elijahshelton/drop-recon>. A modified version of that code to analyse 2D confocal sections of the droplet is available upon request.

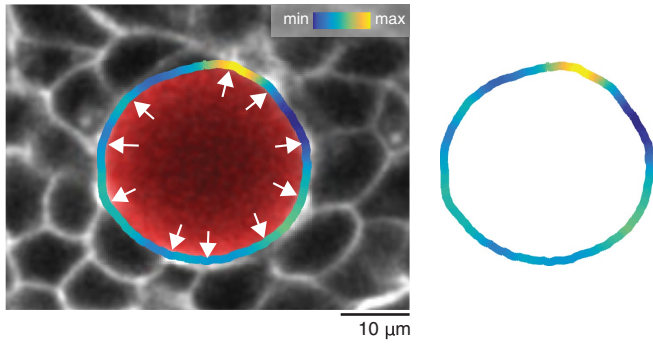
**Data availability.** Source Data for Figs. 1–3 and Extended Data Figs. 1, 5–10 are available in the online version of the paper.

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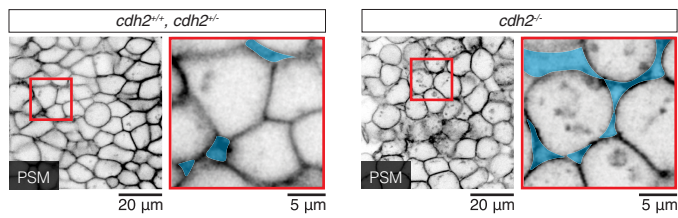


**Extended Data Fig. 1 | Loss of anteroposterior gradients of supracellular stresses and cell and nuclear shape anisotropy in N-cadherin mutants.** **a**, ActinRed staining of F-actin in the PSM of control (*cdh2*<sup>+/+</sup> and *cdh2*<sup>+/-</sup>) and mutant (*cdh2*<sup>-/-</sup>) embryos at the 10-somite stage. Cell shapes are visibly elongated along the mediolateral (ML) direction in control embryos. Cell shape anisotropy is largely lost in *cdh2*<sup>-/-</sup> embryos. **b**, DAPI staining showing higher nuclear mediolateral elongation in the PSM of control embryos compared to *cdh2* mutants. **c**, Frequency of nuclear major axis orientations in the MPZ and PSM (A-PSM and P-PSM). In control (*cdh2*<sup>+/+</sup> and *cdh2*<sup>+/-</sup>) embryos, nuclei in the PSM are elongated along the mediolateral direction, whereas nuclei are oriented randomly in the MPZ. The observed nuclear anisotropy along the mediolateral direction in the PSM of control embryos is decreased in *cdh2* mutants. **d**, A posterior-to-anterior increase in the extent of nuclear elongation (quantified by the nuclear aspect ratio; see inset and Methods) is observed in control (*cdh2*<sup>+/+</sup> and *cdh2*<sup>+/-</sup>) embryos. No anteroposterior gradient in the extent of nuclear elongation (aspect ratio) is observed in *cdh2* mutants (*cdh2*<sup>-/-</sup>). For **c** and **d**: control embryos, *n* = 695 (A-PSM), 752 (P-PSM), 732 (MPZ) nuclei, obtained in 6 embryos per region;

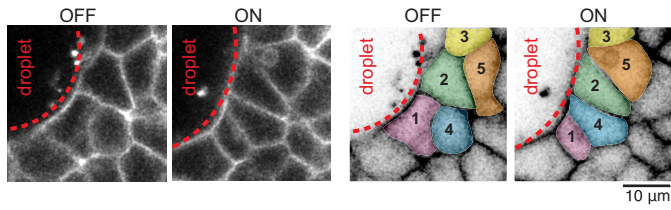
*cdh2*<sup>-/-</sup>: *n* = 833 nuclei from 5 embryos (A-PSM), *n* = 538 nuclei from 6 embryos (P-PSM), *n* = 336 nuclei from 4 embryos (MPZ). Mean ± s.e.m. **e**, Relative change of cell-cell contact length along the anteroposterior axis and the mediolateral axis (*n* = 6,427 and 4,319 cell-cell contacts for PSM and MPZ, respectively, from 5 embryos). Cell junctions are longer along the mediolateral axis compared to the anteroposterior axis, both in the PSM and MPZ. Mean ± s.e.m. **f**, Supracellular stresses are uniform along the anteroposterior axis in *cdh2* mutant embryos (*n* = 5 (A-PSM), 5 (P-PSM), 12 (MPZ)). Mean ± s.e.m.; Mann-Whitney *U*-test. The observed posterior-to-anterior increase in both supracellular stresses and nuclear elongation in control embryos (**d** and Fig. 1e), and the loss of both such gradients in *cdh2* mutants (**d** and **f**), indicate the existence of a N-cadherin-dependent, posterior-to-anterior increase in supracellular stresses, consistent with a posterior-to-anterior increase in mediolateral constriction. Importantly, if the observed thinning of the body axis was caused by pulling forces from the MPZ on the PSM, as previously proposed<sup>18</sup>, both cells and nuclei would be elongated along the anteroposterior axis.



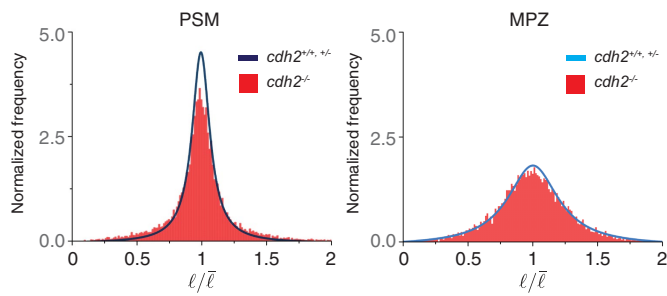
**Extended Data Fig. 2 | Curvature changes along the droplet contour correlate with the locations of cell–cell contacts surrounding the droplet.** Confocal section of a ferrofluid droplet (red) in the MPZ of a *Tg(actb2:MA-citrine)* embryo. The measured curvature values along the detected droplet contour are shown (colourcoded as in Fig. 1h) overlaid with the confocal image (left) and without it (right). White arrows point to locations of cell–cell contacts of cells surrounding the droplet, which correlate with maxima and minima of droplet curvature, consistent with the distance between maxima and minima being approximately the cell size (Fig. 1j).



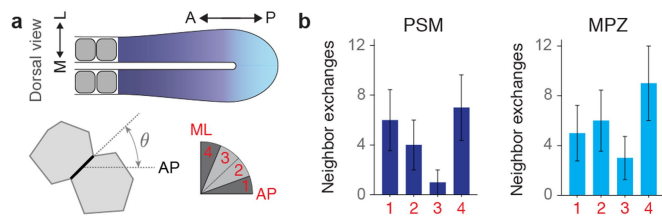
**Extended Data Fig. 3 | Increase of extracellular spaces and cell rounding in *cdh2* mutants.** 2D confocal sections (inverted) of control (*cdh2*<sup>+/+</sup> and *cdh2*<sup>+/-</sup>) and *cdh2*<sup>-/-</sup> Tg(*actb2*:MA-citrine) embryos showing an increase in extracellular space (cyan), as well as more cell rounding, in the PSM tissue of the mutant embryos.



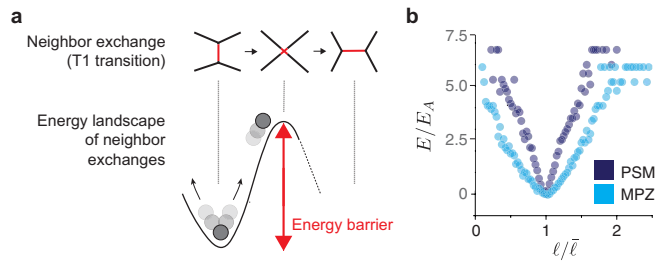
**Extended Data Fig. 4 | Example of neighbour exchanges induced in the tissue upon droplet actuation with a magnetic field.** Confocal section showing the spatial arrangements of cells in the neighbourhood of a magnetically responsive droplet both in the absence of magnetic field (OFF) and after applying a magnetic field (ON) for 15 min (left). Several cell rearrangements are observed to be induced by droplet actuation (right). Some of the cells undergoing neighbour exchanges are coloured and numbered to highlight the rearrangements. Tg(*actb2*:MA-citrine) embryos were used to visualize cell membranes.



**Extended Data Fig. 5 | Distribution of cell–cell contact length fluctuations in *cdh2* mutants.** Normalized frequency (distribution) of cell–cell contact length fluctuations in the PSM and MPZ of *cdh2* mutants (red bars) compared to the control (violet and light blue lines). For PSM and MPZ,  $n = 13,212$  and  $13,634$  cell–cell contacts obtained from 5 and 4 embryos, respectively.



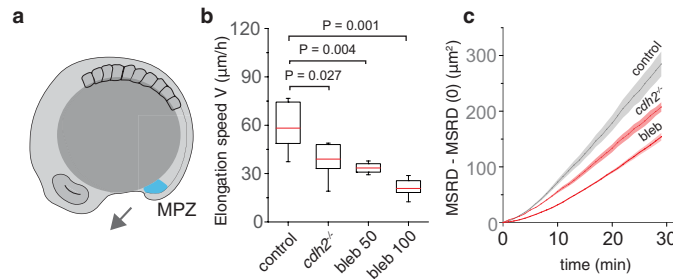
**Extended Data Fig. 6 | Orientation of neighbour exchanges in the MPZ and PSM.** **a**, Sketch of a dorsal view of the elongating body axis, with the anteroposterior and mediolateral directions defined (top). Sketch showing the orientation of a cell-cell contact (thick black line) before undergoing a neighbour exchange (bottom left). The angle  $\theta$  corresponds to the angle between the cell-cell contact before undergoing the neighbour exchange and the anteroposterior axis (bottom). Four equal bins are defined (bin 1:  $0 < \theta < 22.5^\circ$ ; bin 2:  $22.5^\circ < \theta < 45^\circ$ ; bin 3:  $45^\circ < \theta < 67.5^\circ$ ; bin 4:  $67.5^\circ < \theta < 90^\circ$ ) between the anteroposterior and mediolateral orthogonal directions (bottom right). **b**, Frequency of neighbour exchanges along different angular regions ( $n = 18$  in 4 embryos for PSM and  $n = 23$  in 3 embryos for MPZ, with  $n$  being the number of neighbour exchanges analysed). Data are mean  $\pm$  s.d. Neighbour exchanges are largely randomly oriented in the MPZ. In the PSM, neighbour exchanges occur predominantly along either the mediolateral direction or along the anteroposterior axis, with neighbour exchanges occurring slightly less frequently for angles in between these orthogonal orientations. The more frequent occurrence of neighbour exchanges along the anteroposterior and mediolateral axes in the PSM is consistent with the measured directions and extent of ellipsoidal droplet deformation (Fig. 1f), as the persistent and larger supracellular stresses in the PSM may bias neighbour exchanges in these directions. Since neighbour exchanges occur equally frequently along the mediolateral and anteroposterior directions in the PSM, and are uniformly oriented in the MPZ, our results indicate no systematic alignment of neighbour exchanges along a single spatial direction that could potentially contribute to the elongation of the body axis.



### Extended Data Fig. 7 | Energy landscape of neighbour exchanges.

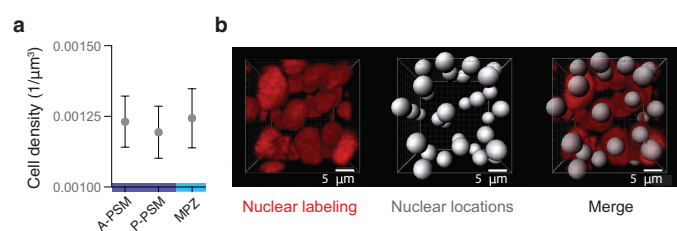
**a**, Schematic of key cellular configurations throughout a neighbour exchange and associated energy landscape. Changing neighbours requires overcoming an energy barrier. Large enough, active cell–cell contact length fluctuations enable neighbour exchanges. **b**, Measured energy

landscape,  $E$ , for PSM and MPZ regions, normalized with the energy scale  $E_A$  associated with cell–cell contact activity or effective temperature energy scale, namely  $E_A = k_B T_{\text{eff}}$ , where  $k_B$  is the Boltzmann constant and  $T_{\text{eff}}$  is the effective temperature.  $n = 6,969, 7,896$  cell–cell contacts obtained from 3, 4 embryos for PSM, MPZ, respectively.

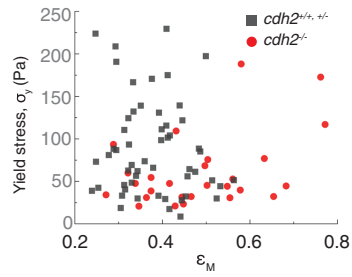


**Extended Data Fig. 8 | Dependence of posterior axis elongation speed and relative cellular movements in the MPZ on N-cadherin and non-muscle myosin-II activity.** **a**, Sketch of a 10-somite stage embryo highlighting the mesodermal progenitor zone (MPZ, cyan) and the direction of posterior elongation (arrow). **b**, Comparison of posterior body elongation speeds between control ( $n = 6$ ),  $cdh2$  mutants ( $n = 7$ ), and blebbistatin-treated embryos ( $n = 6$  for 50  $\mu\text{M}$  and  $n = 7$  for 100  $\mu\text{M}$ ).

Box plots representing median (red line) and second and third quartiles. Error bars indicate 95% confidence interval. Mann–Whitney  $U$ -test. **c**, Mean square relative displacement of cells in the MPZ region of control ( $n = 2,523$  analysed cell pairs from 6 embryos),  $cdh2^{-/-}$  ( $n = 1,154$  analysed cell pairs from 4 embryos) and blebbistatin-treated embryos ( $n = 2,026$  analysed cell pairs from 4 embryos).



**Extended Data Fig. 9 | Cell density is uniform along the anteroposterior axis.** **a**, Measured cell number density (cells per unit volume) in the MPZ, P-PSM and A-PSM. Mean  $\pm$  s.e.m. Cell density does not vary significantly along the anteroposterior axis (within the 10% accuracy of our 3D measurements; Methods). **b**, 3D reconstructions of confocal stacks showing nuclei labelled with H2B::RFP, detected nuclei positions, and composition of both. Cell density was measured using 3D data of nuclear positions in the different regions ( $n = 7,866, 7,214, 11,537$  detected cells in 694, 694, 833 defined boxes in 5, 5, 6 embryos, in A-PSM, P-PSM, MPZ, respectively; Methods).



**Extended Data Fig. 10 | Yield stress values do not depend on the extent of droplet deformation before droplet relaxation.** Measured values of the yield stress plotted against the maximal droplet deformation (maximal applied strain,  $\varepsilon_M$ ; Fig. 2a, b) before starting droplet relaxation. The measured yield stress values do not correlate with the maximal applied strain, neither in control ( $cdh2^{+/+}$  and  $cdh2^{+/-}$ , grey dots,  $n = 53$  embryos) or mutant embryos ( $cdh2^{-/-}$ , red dots,  $n = 27$  embryos). Correlation coefficient,  $r = -0.34$  (control),  $r = -0.04$  ( $cdh2^{-/-}$ ).

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### Statistical parameters

When statistical analyses are reported, confirm that the following items are present in the relevant location (e.g. figure legend, table legend, main text, or Methods section).

n/a Confirmed

- ☐ ☒ The exact sample size ( $n$ ) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ An indication of whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided  
*Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
- ☒ ☐ A description of all covariates tested
- ☒ ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistics including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g.  $F$ ,  $t$ ,  $r$ ) with confidence intervals, effect sizes, degrees of freedom and  $P$  value noted  
*Give  $P$  values as exact values whenever suitable.*
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
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- ☒ ☐ Estimates of effect sizes (e.g. Cohen's  $d$ , Pearson's  $r$ ), indicating how they were calculated
- ☐ ☒ Clearly defined error bars  
*State explicitly what error bars represent (e.g. SD, SE, CI)*

Our web collection on [statistics for biologists](#) may be useful.

### Software and code

Policy information about [availability of computer code](#)

Data collection

Commercial Zeiss Zen software was used to perform confocal imaging. Software to perform magnetic actuation of droplets was previously published in Ref. 20.

Data analysis

Commercial software used to analyze data: Imaris 8.0 (Bitplane) and Matlab 9.2 (MathWorks). ImageJ 1.51k was also used to analyze data. Commercial software COMSOL Mutiphysics 5.3 was used to perform simulations. Other software: Tissue Analyzer (previously published; ref. 37; publicly available). Custom software to analyze droplet deformations was also used and previously described in ref. 35. This custom code is publicly available on GitHub.

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## Data

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All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
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- A description of any restrictions on data availability

Source data supporting these findings are available online as Supplementary Data.

## Field-specific reporting

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## Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

|                 |                                                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sample size     | Sample size was chosen so that statistical variation of the data did not change considerably upon addition of more data points.                                                 |
| Data exclusions | No data was excluded.                                                                                                                                                           |
| Replication     | Measurements were independently repeated to confirm that different data sets of the same quantity did not change the reported results.                                          |
| Randomization   | No specific experimental groups were defined and all data was considered.                                                                                                       |
| Blinding        | Analysis was done by automated software which was blind to data collection. Team members analyzing the data did not know about the origin of the data until after the analysis. |

## Reporting for specific materials, systems and methods

### Materials & experimental systems

| n/a                                 | Involved in the study                                           |
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| <input type="checkbox"/>            | <input checked="" type="checkbox"/> Unique biological materials |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> Antibodies                             |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> Eukaryotic cell lines                  |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> Palaeontology                          |
| <input type="checkbox"/>            | <input checked="" type="checkbox"/> Animals and other organisms |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> Human research participants            |

### Methods

| n/a                                 | Involved in the study                           |
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| <input checked="" type="checkbox"/> | <input type="checkbox"/> Flow cytometry         |
| <input checked="" type="checkbox"/> | <input type="checkbox"/> MRI-based neuroimaging |

## Unique biological materials

Policy information about [availability of materials](#)

Obtaining unique materials All materials used are commercially available, with the exception of the fluorocarbon Rhodamine dye, which was a kind gift from Dr. Ellen Sletten (UCLA). Details on the synthesis of this fluorocarbon Rhodamine dye can be found in ref. 33.

## Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals Zebrafish (*Danio rerio*) were used in this study. Since only embryos were studied, sex-specific experiments were not necessary, as zebrafish embryos at the studies stage have not yet undergone sex determination.

Wild animals

None.

Field-collected samples

None.